

ABSTRACT

CampusPulse Recruitment Automation System is a full-stack, AI-enhanced web application designed to eliminate manual chaos, scheduling conflicts, and communication gaps in college placement drives. Serving Students, Recruiters, and TPOs through role-based dashboards, the system introduces a **Semantic Skill-Intelligence Engine** powered by Sentence-BERT embeddings that ranks candidates based on the contextual meaning of resumes against job descriptions, moving beyond basic keyword matching. A **Verification Pipeline** auto-blocks ineligible applicants based on CGPA and backlogs, while a **Conflict-Detection Algorithm** using MongoDB aggregation prevents real-time double-bookings across Aptitude Tests, Group Discussions, and Interviews.

The platform supports a **Multi-Stage Recruitment Pipeline**—recruiters can schedule tests, group discussions, and interviews, with application statuses updating dynamically across the entire funnel. Additional features include **CSV Bulk Import** for student data, a **Progressive Web App (PWA)** for offline schedule access, **Automated Email Reminders** via node-cron and Nodemailer, **Recruiter Trust Score** verification using WHOIS and MCA links, and an **ECharts-based TPO Dashboard** for placement analytics with stage-wise funnel visualization.

Built as a monolithic Node.js application with a Python microservice for the AI engine, deployed locally on a single server for a single institution, this Mini Project delivers a fully functional, transparent, and efficient automation prototype—proving the core workflow before scaling to enterprise-grade architecture in the Main Project.

Modules

1. Student Module

- **Login** using Roll Number and password (accounts are pre-created by TPO via CSV; students activate via email link).
- **Complete Profile** (upload resume PDF, enter CGPA, branch, passout year, active backlogs).
- View a personalized dashboard listing only the drives they are eligible for (based on CGPA/branch/backlog verification).
- **Apply to drives** with a single click.
- Track their **Application Status** in real-time through the complete multi-stage pipeline:

- *Applied → Under Review → Aptitude Scheduled → Aptitude Completed → GD Scheduled → GD Completed → Interview Scheduled → Interview Completed → Selected/Rejected/Waitlisted → Offer Accepted/Declined → Placed.*
 - *(Stages are dynamic—if a recruiter does not include Aptitude/GD, the system automatically skips those stages.)*
- **Receive Explainable AI (XAI) Feedback**—when shortlisted or rejected at the resume stage, they see specific sentences from their resume that positively or negatively influenced their AI match score, ensuring complete transparency.
- **Receive Live System Alerts** (pop-up notifications) when a recruiter schedules their test, GD, interview, or updates their status.
- **View Offer Letters**—when selected, view and download the official offer letter (PDF) uploaded by the recruiter.
- **Accept or Decline Offers** directly from their dashboard with a single click; declination reasons are recorded for TPO analytics.
- **View Upcoming Schedule** (test date/time, GD slot, interview date/time) with automated 24-hour email reminders and .ics calendar invites for each stage.
- **Access as a Progressive Web App (PWA)**—install on phone and view schedules offline.

2. Company Recruiter Module

- **Register their company** (awaiting TPO approval).
- **Complete Company Profile** (official email, GST, website, company logo).
- **Post Job Drives** with detailed criteria (CTC, job description, application deadlines, minimum CGPA, eligible branches, and backlog limits).
- **Configure Multi-Stage Pipeline**—toggle options to include Aptitude Test and/or Group Discussion for the drive. If toggled off, students move directly to the interview stage after shortlisting.
- **View All Applications** received for their posted drives with AI-generated match scores.

- **View Explainable AI (XAI) Justifications**—for each candidate, see *why* they were ranked high or low (specific skills/experiences that contributed to the score).
- **Shortlist or Reject Applicants** (with system-assistance from the Semantic Ranker).
- **Schedule Aptitude Test**—set date/time and upload test link/instructions for shortlisted candidates. The Conflict-Detection Algorithm validates against students' existing commitments and blocks overlaps in real-time.
- **Mark Aptitude Results**—update student statuses to *Pass* or *Fail* in bulk or individually. Students who pass automatically move to the next stage (GD or Interview).
- **Schedule Group Discussion**—assign GD slots to students who passed the aptitude test. Conflict-Detection Algorithm ensures no overlaps with other scheduled activities.
- **Mark GD Results**—update student statuses to *Shortlisted* or *Rejected* in bulk or individually. Shortlisted students automatically move to the interview stage.
- **Manually Schedule Interviews**—assign interview slots to shortlisted students. Conflict-Detection Algorithm validates against all existing commitments and blocks overlaps in real-time.
- **Declare Final Results** after interviews (Select/Reject/Waitlist) in bulk or individually.
- **Upload Official Offer Letters (PDF)** for selected students—the system auto-triggers a congratulatory email with a personalized download link.
- **View Offer Acceptance Rates**—track how many selected students accepted vs. declined their offers.
- **Recruiter Trust Score Assistance**—the system displays WHOIS domain age and provides direct links to MCA/Glassdoor (final verification by TPO).

3. TPO (Admin) Module

- **Login** with full administrative privileges.
- **Approve or Reject** newly registered Company Recruiters after reviewing their Trust Score (WHOIS/MCA data).
- **CSV Bulk Import**—upload an Excel/CSV file to instantly migrate hundreds of student records (name, email, CGPA, branch, backlogs, etc.). The system auto-triggers activation emails to students.
- **View Advanced ECharts Dashboard** showing:
 - Total placed vs. unplaced students.
 - Branch-wise placement percentages.
 - **Stage-wise Funnel Visualization**—track candidate progression: Applications Received → Aptitude Cleared → GD Cleared → Interviewed → Placed.
 - **Offer-to-Acceptance Ratio** (how many offers were accepted vs. declined, with trend analysis).
 - **Stage-wise Drop-off Analytics**—identify at which stage (Aptitude/GD/Interview) the maximum number of students are filtered out.
 - Company-wise hiring trends and historical performance.
- **Oversee All Active Drives** across all companies with real-time status tracking at every stage.